Q.20 Solve the following Linear Programming Problems graphically:

Maximise $z=3x+4y$ subject to the constraints: $x+y \leq 4,\ x \geq 0,\ y \geq 0$.

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Roll No. ....

DVOC (Level 3)

2nd Sem. / DVOC (Ref. & Air Cond., Auto. Servicing, ITM, PT, SD, AMT, FP, EMS, GM)

Subject : Applied Mathematics-II

Time : 2 Hrs. M.M. : 50

Section-A

Note: Multiple Choice questions. Attempt all questions. (10x1=10)

Q.1 Write an example of square matrix.

Q.2 Evaluate $\begin{vmatrix} 4 & 2 \\ 1 & 3 \end{vmatrix}$

Q.3 If $f(x)= x^2+3x+2$. Find $f(1)$.

Q.4 The value of $\lim_{x\to 0}\frac{\sin x}{x}$ is

Q.5 $\frac{d}{dx}(\sec x) =$ __________

Q.6 $\int e^{2x}dx =$ ______

Q.7 $\int_0^1 x\,dx =$ ______

Q.8 The order of the differential equation: $\frac{d^2y}{dx^2} + 2y = 3\tan x$ is ______.

Q.9 Write the slope of the straight line $3x+2y+5=0$

Q.10 Define Unit Vector.

Section-B

Note: Short answer type questions. Attempt any six questions out of eight questions. (6x5=30)

Q.11 If $A = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$ then find 3A+2B.

Q.12 Evaluate $\lim_{x\to 2} \frac{x^2 - 4}{x - 2}$

Q.13 Differentiate $y=x^2 \cos x$ with respect to x.

Q.14 Evaluate the product: $(3\vec{a} - 5\vec{b}).(2\vec{a} + 7\vec{b})$

Q.15 Evaluate $\int \left(x^3 + 5sinx + e^x + 1\right) dx$

Q.16 Evaluate $\int_1^2 \left(x^2 + 1\right) dx$

Q.17 Find the angle between the lines whose slopes are 3 and $\frac{1}{2}$.

Q.18 Solve the differential equation: $\frac{dy}{dx} = x^2 + 3x + 2.$

Section-C

Note: Long answer type questions. Attempt any one question out of two questions. (1x10=10)

Q.19 Find the all points of maxima and minima and their corresponding maximum and minimum value of the function $y=x^3-6x^2+9x-9$.